

Higher Order Linear ODEs: Variable coefficients

Cauchy-Euler equation:

A linear differential equation of the form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_2 x^2 \frac{d^2 y}{dx^2} + a_1 x \frac{dy}{dx} + a_0 y = g(x)$$

where $a_n, a_{n-1}, a_{n-2}, \dots, a_0$ are constants.

Second order: Solving through trial solution

Example. $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = 0$

Solution. Let, $y = x^m$ is trial solution of the given homogeneous ODE. Then we have,

$$x^2 \cdot m(m-1)x^{m-2} - 2x \cdot mx^{m-1} - 4x^m = 0$$

$$\Rightarrow \{m(m-1) - 2m - 4\}x^m = 0$$

$$\Rightarrow m(m-1) - 2m - 4 = 0 \quad [\because x^m \neq 0]$$

$$\Rightarrow m^2 - 3m - 4 = 0 \Rightarrow m = -1, 4$$

The general solution of the ODE becomes,

$$y = c_1 x^{-1} + c_2 x^4$$

For real and distinct roots m_1 and m_2 :

$$y = c_1 x^{m_1} + c_2 x^{m_2}$$

Higher Order Linear ODEs: Variable coefficients

Second order homogeneous form: Solving through trial solution

Example. $4x^2 \frac{d^2 y}{dx^2} + 8x \frac{dy}{dx} + y = 0$

Solution. Let, $y = x^m$ is trial solution of the given homogeneous ODE. Then we have,

$$4x^2 \cdot m(m-1)x^{m-2} + 8x \cdot mx^{m-1} + x^m = 0$$

$$\Rightarrow \{4m(m-1) + 8m + 1\}x^m = 0$$

$$\Rightarrow 4m(m-1) + 8m + 1 = 0 \quad [\because x^m \neq 0]$$

$$\Rightarrow 4m^2 + 4m + 1 = 0$$

$$\Rightarrow (2m+1)^2 = 0 \Rightarrow m = -\frac{1}{2}, -\frac{1}{2}$$

The general solution of the ODE becomes,

$$y = (c_1 + c_2 \ln x)x^{-\frac{1}{2}}$$

For real and equal roots m_1 and m_1 :
 $y = (c_1 + c_2 \ln x)x^{m_1}$

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Second order homogeneous form: Solving through change of variables

Example. $x^2 \frac{d^2 y}{dx^2} - 5x \frac{dy}{dx} + 9y = 0$

Solution. Let, $t = \ln x \Rightarrow x = e^t$, and $y(x) = z(\ln x) = z(t)$

Here, $\frac{dy}{dx} = \frac{d}{dx}(z(t)) = \frac{dz}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \frac{dz}{dt} \Rightarrow x \frac{dy}{dx} = \frac{dz}{dt}$

Similarly, $\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{x} \frac{dz}{dt} \right) = \frac{1}{x} \frac{d}{dx} \left(\frac{dz}{dt} \right) - \frac{1}{x^2} \frac{dz}{dt} = \frac{1}{x^2} \frac{d^2 z}{dt^2} - \frac{1}{x^2} \frac{dz}{dt} \Rightarrow x^2 \frac{d^2 y}{dx^2} = \frac{d^2 z}{dt^2} - \frac{dz}{dt}$

The Cauchy-Euler differential equation is transformed into, $\frac{d^2 z}{dt^2} - 6 \frac{dz}{dt} + 9z = 0$

which is a 2nd order homogeneous ODE with constant coefficients and its auxiliary equation becomes, $m^2 - 6m + 9 = 0 \Rightarrow (m - 3)^2 = 0 \Rightarrow m = 3, 3$

The general solution of the ODE becomes,

$$z(t) = (c_1 + c_2 t)e^{3t} \text{ and hence, } y(x) = (c_1 + c_2 \ln x)x^3$$

For real and equal roots m_1 and m_1 :
 $y = (c_1 + c_2 \ln x)x^{m_1}$

Higher Order Linear ODEs: Variable coefficients

Second order homogeneous form: Solving through trial solution

Example. $4x^2 \frac{d^2 y}{dx^2} + 17y = 0$, $y(1) = -1$, $y'(1) = -\frac{1}{2}$

Solution. Let, $y = x^m$ is trial solution of the given homogeneous ODE. Then we have,

$$4x^2 \cdot m(m-1)x^{m-2} + 17x^m = 0$$

$$\Rightarrow \{4m(m-1) + 17\}x^m = 0$$

$$\Rightarrow 4m(m-1) + 17 = 0 \quad [\because x^m \neq 0]$$

$$\Rightarrow 4m^2 - 4m + 17 = 0$$

$$\Rightarrow (2m-1)^2 = -16 \Rightarrow m = \frac{1}{2} \pm 2i$$

$$\text{For complex roots } m_1 \pm im_2: \\ y_c = x^{m_1} [c_1 \cos(m_2 \ln x) + c_2 \sin(m_2 \ln x)]$$

The general solution of the ODE becomes,

$$y = x^{\frac{1}{2}} [c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x)]$$

Applying the boundary conditions, $y(1) = -1$, $y'(1) = -\frac{1}{2}$, we obtain $c_1 = -1$ and $c_2 = 0$

The required solution of the given IVP, $y = -x^{\frac{1}{2}} \cos(2 \ln x)$.

Higher Order Linear ODEs: Variable coefficients

Second order homogeneous form: Solving through trial solution

Example. $x^3 \frac{d^3 y}{dx^3} + 5x^2 \frac{d^2 y}{dx^2} + 7x \frac{dy}{dx} + 8y = 0$

Solution. Let, $y = x^m$ is trial solution of the given homogeneous ODE. Then we have,

$$x^3 \cdot m(m-1)(m-2)x^{m-3} + 5x^2 \cdot m(m-1)x^{m-2} + 7x \cdot mx^{m-1} + 8x^m = 0$$

$$\Rightarrow \{m(m-1)(m-2) + 5m(m-1) + 7m + 8\}x^m = 0$$

$$\Rightarrow m(m-1)(m-2) + 5m(m-1) + 7m + 8 = 0 \quad [\because x^m \neq 0]$$

$$\Rightarrow m^3 + 2m^2 + 4m + 8 = 0$$

$$\Rightarrow m^2(m+2) + 4(m+2) = 0 \Rightarrow (m+2)(m^2+4) = 0 \Rightarrow m = -2, \pm 2i$$

The general solution of the ODE becomes,

$$y = c_1 x^{-2} + c_2 \cos(2 \ln x) + c_3 \sin(2 \ln x)$$

Higher Order Linear ODEs: Variable coefficients

Second order non-homogeneous form: Solving through trial solution

Example. $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 3y = 2x^4 e^x$

Solution. Let, $y = x^m$ is trial solution of the homogeneous form of given ODE.

Then we have,

$$x^2 \cdot m(m-1)x^{m-2} - 3x \cdot mx^{m-1} + 3x^m = 0$$

$$\Rightarrow \{m(m-1) - 3m + 3\}x^m = 0 \Rightarrow m(m-1) - 3m + 3 = 0 \quad [\because x^m \neq 0]$$

$$\Rightarrow m^2 - 4m + 3 = 0 \Rightarrow (m-1)(m-3) = 0 \Rightarrow m = 1, 3$$

The complementary solution of the ODE becomes, $y_c = c_1 x + c_2 x^3$

The standard form of the given ODE becomes, $\frac{d^2 y}{dx^2} - \frac{3}{x} \frac{dy}{dx} + 3 \frac{y}{x^2} = 2x^2 e^x$

for which the particular solution can be written as, $y_p = A(x) x + B(x) x^3$

Here, the **Wronskian** of the solutions $y_1 = x$ and $y_2 = x^3$ yields,

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} x & x^3 \\ 1 & 3x^2 \end{vmatrix} = 3x^3 - x^3 = 2x^3 \neq 0$$

Higher Order Linear ODEs: Variable coefficients

Second order non-homogeneous form: Solving through trial solution

Example. $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 3y = 2x^4 e^x \Rightarrow \frac{d^2 y}{dx^2} - \frac{3}{x} \frac{dy}{dx} + 3 \frac{y}{x^2} = 2x^2 e^x$

Solution. Therefore, the solutions y_1 and y_2 are linearly independent. We also compute,

$$W_1(y_1, y_2) = \begin{vmatrix} 0 & y_2 \\ R(x) & y_2' \end{vmatrix} = \begin{vmatrix} 0 & x^3 \\ 2x^2 e^x & 3x^2 \end{vmatrix} = -2x^5 e^x,$$

$$W_2(y_1, y_2) = \begin{vmatrix} y_1 & 0 \\ y_1' & R(x) \end{vmatrix} = \begin{vmatrix} x & 0 \\ 1 & 2x^2 e^x \end{vmatrix} = 2x^3 e^x.$$

Now we find,

$$A(x) = \int \frac{W_1}{W} dx = - \int \frac{2x^5 e^x}{2x^3} dx = - \int x^2 e^x dx = -(x^2 e^x - 2x e^x + 2e^x),$$

$$B(x) = \int \frac{W_2}{W} dx = \int \frac{2x^3 e^x}{2x^3} dx = \int e^x dx = e^x.$$

Thus the particular solution becomes, $y_p = -(x^2 e^x - 2x e^x + 2e^x) x + (e^x) x^3$
 $\Rightarrow y_p = 2x^2 e^x - 2x e^x$

Finally, the general solution of the given problem yields,

$$y = y_c + y_p = c_1 x + c_2 x^3 + 2x^2 e^x - 2x e^x.$$

Higher Order Linear ODEs: Variable coefficients

Second order non-homogeneous form: Solving through change of variables

Example. $xy'' + y' = x$

Solution. . Let, $t = \ln x \Rightarrow x = e^t$, and $y(x) = z(\ln x) = z(t)$

$$\text{Here, } \frac{dy}{dx} = \frac{d}{dx}(z(t)) = \frac{dz}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \frac{dz}{dt} \Rightarrow x \frac{dy}{dx} = \frac{dz}{dt}$$

$$\text{Similarly, } \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x} \frac{dz}{dt}\right) = \frac{1}{x} \frac{d}{dx}\left(\frac{dz}{dt}\right) - \frac{1}{x^2} \frac{dz}{dt} = \frac{1}{x^2} \frac{d^2z}{dt^2} - \frac{1}{x^2} \frac{dz}{dt} \Rightarrow x^2 \frac{d^2y}{dx^2} = \frac{d^2z}{dt^2} - \frac{dz}{dt}$$

Rewriting the given differential equation into, $xy'' + y' = x \Rightarrow x^2y'' + xy' = x^2$

$$\text{and then transforming into, } \frac{d^2z}{dt^2} - \frac{dz}{dt} + \frac{dz}{dt} = e^{2t} \Rightarrow \frac{d^2z}{dt^2} = e^{2t}$$

which is a 2nd order homogeneous ODE with constant coefficients and its auxiliary equation becomes, $m^2 = 0 \Rightarrow m = 0, 0$

The complementary solution of the ODE becomes, $z(t) = c_1 + c_2t \Rightarrow y_c(x) = c_1 + c_2 \ln x$

And the particular solution becomes, $z_p(t) = \frac{1}{D^2} e^{2t} = \frac{1}{4} e^{2t} \Rightarrow y_p(x) = \frac{1}{4} x^2$

The general solution yields, $y(x) = y_p(x) + y_c(x) = c_1 + c_2 \ln x + \frac{1}{4} x^2$

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Exercise 4.7

Solve the following differential equations

1. $x^2y'' - 2y = 0$

2. $4x^2y'' + y = 0$

3. $xy'' + y' = 0$

4. $xy'' - 3y' = 0$

5. $x^2y'' + xy' + 4y = 0$

6. $x^2y'' + 5xy' + 3y = 0$

7. $x^2y'' - 3xy' - 2y = 0$

8. $x^2y'' + 3xy' - 4y = 0$

9. $25x^2y'' + 25xy' + y = 0$

10. $4x^2y'' + 4xy' - y = 0$

11. $x^2y'' + 5xy' + 4y = 0$

12. $x^2y'' + 8xy' + 6y = 0$

13. $3x^2y'' + 6xy' + y = 0$

14. $x^2y'' - 7xy' + 41y = 0$

15. $x^3y''' - 6y = 0$

16. $x^3y''' + xy' - y = 0$

17. $xy^{(4)} + 6y''' = 0$

18. $x^4y^{(4)} + 6x^3y''' + 9x^2y'' + 3xy' + y = 0$

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Exercise 4.7

Solve the following differential equations

19. $xy'' - 4y' = x^4$

20. $2x^2y'' + 5xy' + y = x^2 - x$

21. $x^2y'' - xy' + y = 2x$ 22. $x^2y'' - 2xy' + 2y = x^4e^x$

23. $x^2y'' + xy' - y = \ln x$ 24. $x^2y'' + xy' - y = \frac{1}{x+1}$

25. $x^2y'' + 3xy' = 0, \quad y(1) = 0, y'(1) = 4$

26. $x^2y'' - 5xy' + 8y = 0, \quad y(2) = 32, y'(2) = 0$

27. $x^2y'' + xy' + y = 0, \quad y(1) = 1, y'(1) = 2$

28. $x^2y'' - 3xy' + 4y = 0, \quad y(1) = 5, y'(1) = 3$

29. $xy'' + y' = x, \quad y(1) = 1, y'(1) = -\frac{1}{2}$

30. $x^2y'' - 5xy' + 8y = 8x^6, \quad y\left(\frac{1}{2}\right) = 0, y'\left(\frac{1}{2}\right) = 0$